

HW 7

MSE

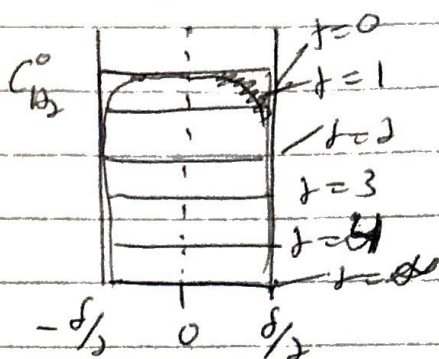
360

Guse 3/6/05

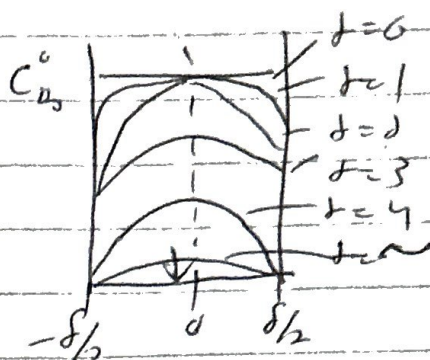
1) attached in Matlab

2)

(b)



(c)



a) most likely, the diffusion of H_2 through steel will be the RLS, since leaving the surface is not energetically difficult.

b) (drawn above)

c) (drawn above)

d) finite

e) $C(x=0, t=0) = C_{H_2}^0$

$C(x=\delta/2, t) = 0$

$C(x=-\delta/2, t) = 0$

$\delta = 4 \text{ mm}$

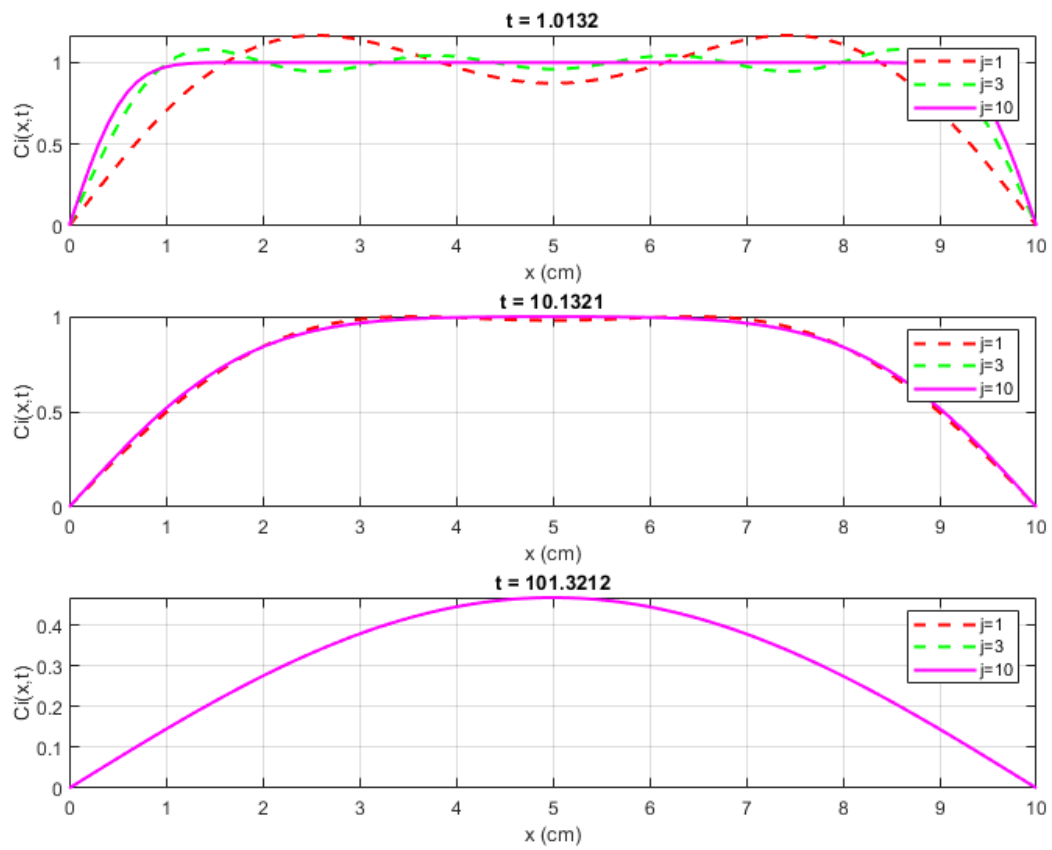
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Cstar = 0; % given
Ci0 = 1; % random C0
L = 10; % random thickness
D = 0.1;
x_vals = linspace(0, L, 100);
t1 = 0.01*(L^2/(pi()^2*D));
t2 = 0.1*(L^2/(pi()^2*D));
t3 = 1*(L^2/(pi()^2*D));
times = [t1, t2, t3];
figure('Position', [100 100 800 600]);
titles = {'t = ', num2str(t1)}, ['t = ', num2str(t2)], ['t = ',
num2str(t3)]};
Ci_function = @(x, t, j_max) Cstar + (Ci0 - Cstar) * (4/pi) * ...
    sum(arrayfun(@(j) (1/(2*j + 1)) * sin((2*j + 1) * pi * x / L) * ...
        exp(-(2*j + 1)^2 * pi^2 * D * t) / L^2), 0:j_max));

% plot
for k = 1:length(times)
    t = times(k);
    subplot(3,1,k);
    Ci_j1 = arrayfun(@(x) Ci_function(x, t, 1), x_vals);
    Ci_j3 = arrayfun(@(x) Ci_function(x, t, 3), x_vals);
    Ci_j10 = arrayfun(@(x) Ci_function(x, t, 10), x_vals);
    plot(x_vals, Ci_j1, 'r--', 'LineWidth', 1.5);
    hold on;
    plot(x_vals, Ci_j3, 'g--', 'LineWidth', 1.5);
    plot(x_vals, Ci_j10, 'm-', 'LineWidth', 1.5);
    hold off;
    xlabel('x (cm)');
    ylabel('Ci(x,t)');
    title(titles{k});
    legend('j=1', 'j=3', 'j=10');
    grid on;
end

% Comment on results: its clear that a higher J term is needed at lower
% times to approximate it. it seems that past t2, a single j term is
% acceptable.

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